

WEEKLY UPDATE 1

CSE499B — Senior Design Project | Week of February 24, 2026

1. Work Accomplished This Week

- Successfully exported the BLIP-based vision encoder and text decoder models to ONNX format (~900 MB total), resolving BLIP-2 incompatibility caused by unsupported float16 operations on Android.
- Integrated ONNX Runtime v1.20.0 into the Android project and resolved a model IR version mismatch (IR v10 vs. max supported v9) by upgrading the runtime dependency.
- Debugged and corrected multiple Kotlin type mismatch errors in the inference pipeline (Int vs. Long tensor shapes, named parameter API misuse in `OnnxTensor.createTensor`).
- Deploy all required model files to the test device (Xiaomi 2201116SI) via ADB, including the large external `.onnx.data` weight files that were initially missing.
- Achieved fully functional end-to-end on-device image captioning — the app correctly generated the caption "a laptop computer sitting on top of a desk" from a live camera image with no internet connection.

2. Planned Work for Next Week

- Apply INT8 dynamic quantization to reduce model storage from ~900 MB to an estimated ~224 MB (approximately 75% reduction).
- Benchmark inference latency and memory usage on the test device and document results.
- Conduct caption accuracy evaluation across diverse image categories.
- Refine the application of UI for project demonstration readiness.

3. Overall project milestone

1. Model Export (BLIP-base ONNX) — Complete

Vision encoder + text decoder exported to ONNX with all required tokenizer files.

2. Android App & Camera Integration — Complete

CameraX + ONNX Runtime v1.20.0 fully integrated; app builds and installs on Xiaomi device.

3. On-Device Inference & Caption Generation — Complete

End-to-end pipeline confirmed working; accurate natural language captions generated offline.

4. Model Quantization (INT8) — In Progress

Dynamic quantization to be applied next week to reduce model size and improve inference speed.

5. Benchmarking & Final Documentation — Pending

Latency, memory profiling, and final report write-up scheduled for project conclusion.